

Of linear models & beyond; Supervised learning

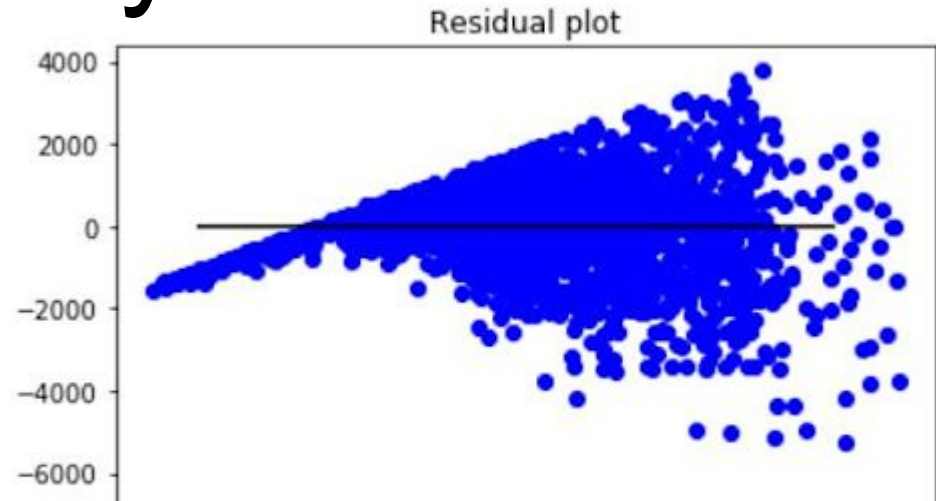
Victor Gimenez-Abalos

Linear model hypotheses used

- Homoscedasticity: y 's variance is equal for all data points.
 - Is this often the case in reality? Is it always the case?
- Residuals come from a gaussian (Output is any in $\mathbb{R}$)
 - What if ' y ' is a probability of something? How do residuals truly look?
- **(CRIPPLING)** Linear relationship input-output
 - What if $y=x^2$?

Let's break these assumptions

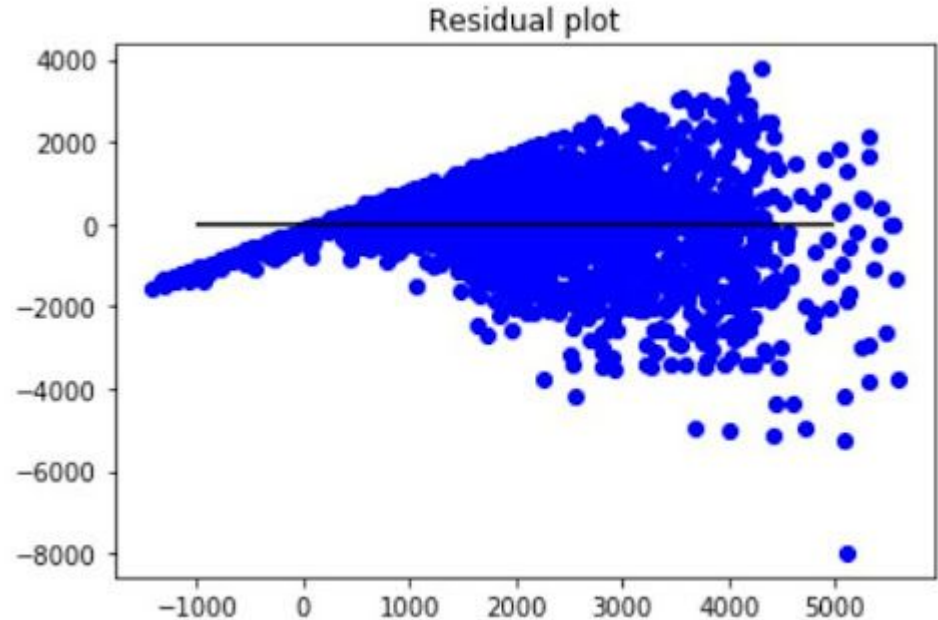
Homoscedasticity



Homoscedasticity reminder

- Typical example: predict savings from income
 - Low income people have likely low savings, proportional
 - As income increases, there **may** be many people who don't have savings.

Very typical in problems where there are **bounds to y**



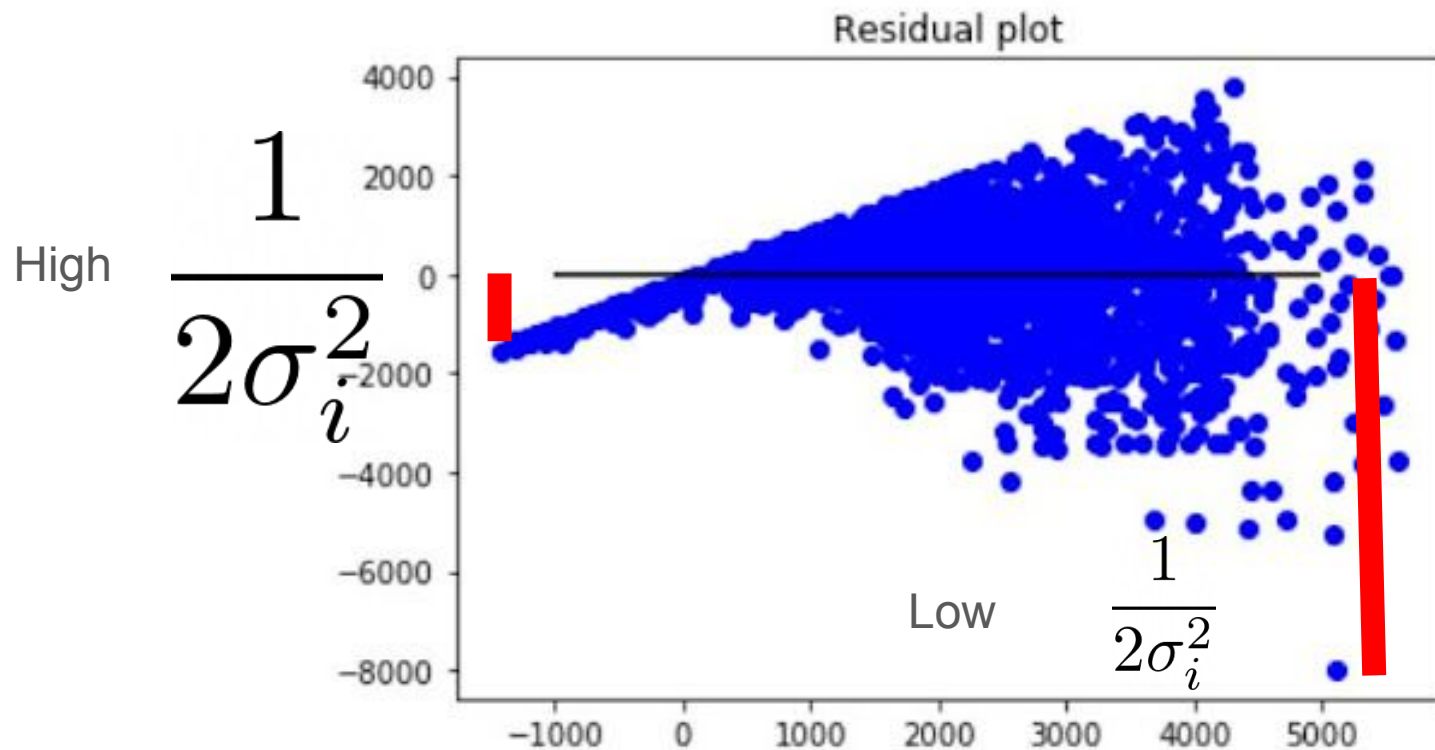
Breaking Homoscedasticity (WLS)

- What if heteroscedasticity? How did we get to MSE:

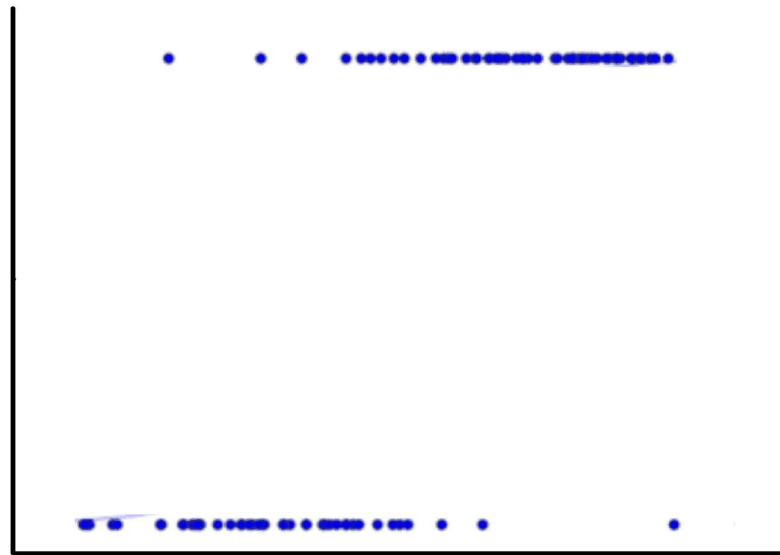
$$\begin{aligned} P(D|M) &= \prod_i P(d_i|M) = \prod_i P(y_i|x_i, M)P(x_i) \\ \log P(D|M) - \sum_i \log P(X_i) &= \sum_i \log P(y_i|x_i, M) = - \sum_i \frac{1}{2} \log 2\pi\sigma^2 - \sum_i \frac{(y_i - M(x_i))^2}{2\sigma^2} \\ &\quad - \sum_i \frac{(y_i - M(x_i))^2}{2\sigma_i^2} \end{aligned}$$

Weighted Least Squares method. You need variance of each sample. There are many proposals on how to do it. Idea: Bagging!

Visually

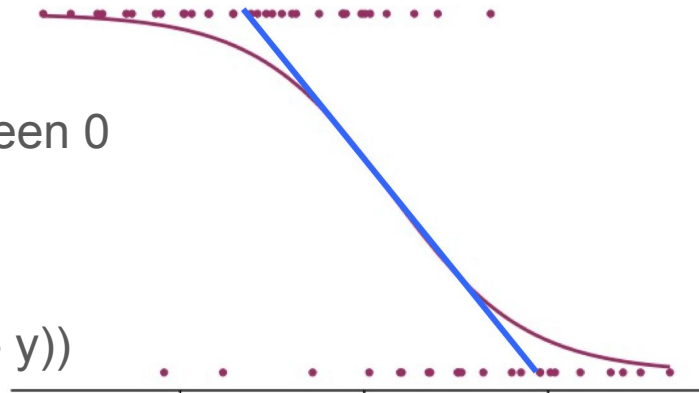


Output Distribution



Breaking Output Distribution (GLMs)

- In real world problems, some outputs are more probable than others.
 - Eg: What if we know output is always between 0 and 1? Then error is not gaussian!
- GLMs: idea is to predict **$g(y)$** , not y .
 - Eg, if y is between 0 and 1; $g(y) = \ln(y / (1 - y))$ (Inverse of logistic function!).
 - Train f with $(x, g(y))$ as always
 - Inference as $g^{-1}(f(x))$



Which g?

First, choose/guess output distribution, and find the linked 'g'

- Normal? $g(y)=y$ -> Linear regression (as before!)
- Exponential? Gamma? $g(y) = -1/y$
- Bernoulli/Binomial/...? **Logit**: $g(y)=\log(y/(n-y))$; ($n=1$ for all except binomial)

Advantage: prediction is more informed.

Disadvantage: you have to inform the prediction

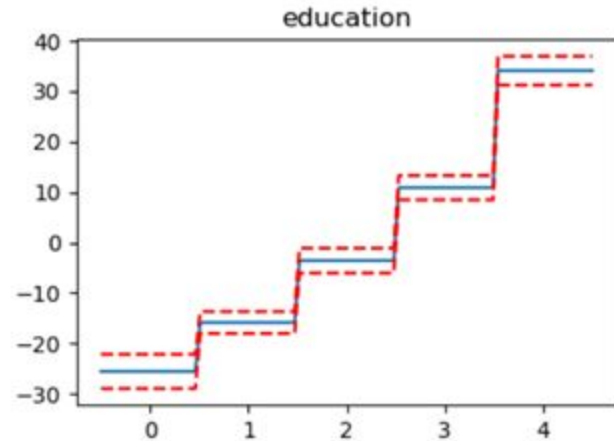
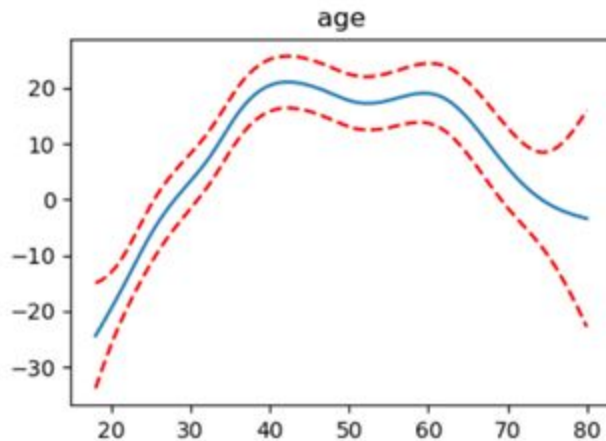
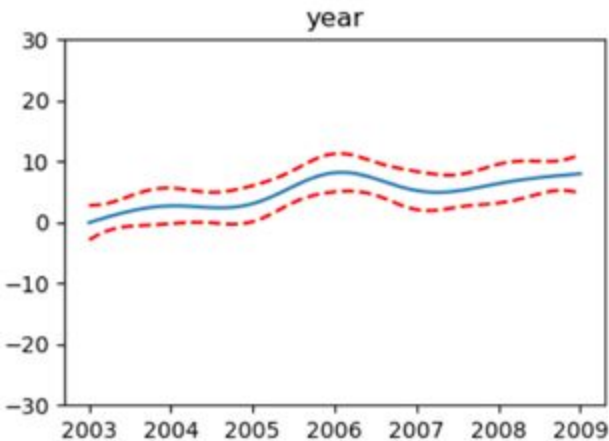
Reality: you should always inform your predictions (inductive biasing)!

Linearity & Expressivity

Expressivity

What if the relationship between variables and targets is **not** linear?

Income problem:

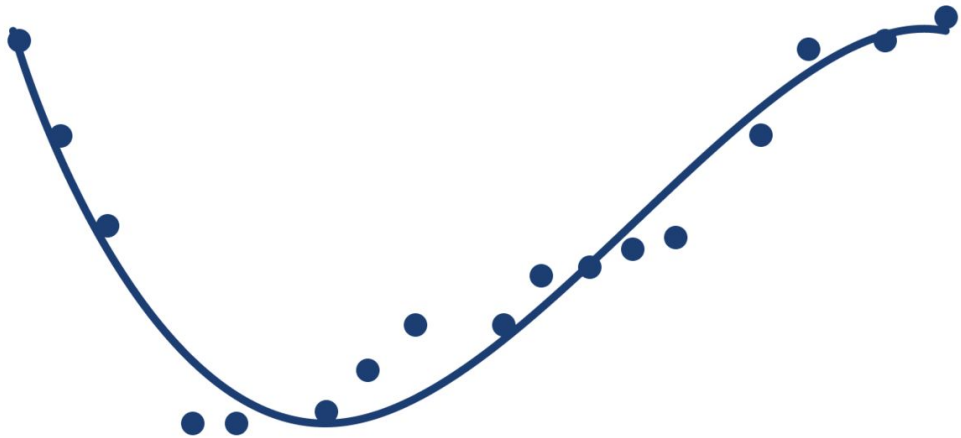


Cheeky idea: do the same

Kernel method: expand variables by adding transformations! Which ones? Pick yourself, **make a design matrix**. No further trick is needed, train with expanded variables.

Polynomial regression, Universal Approximation.

What happens with
the Bias Variance tradeoff?



Slightly different ideas also possible

- GAMs: Generalised Additive Models

$$g(y) = w_0 + f_1(x_1) + f_2(x_2) + \dots f_m(x_m)$$

- GAIMs: Interactive (variable interactions)

$$g(y) = w_0 + f_1(x_1) + f_{12}(x_1, x_2) + f_2(x_2) + \dots f_m(x_m)$$

Problem: where/how do we get these functions?

- This is a **family** of methods with thousands of variants

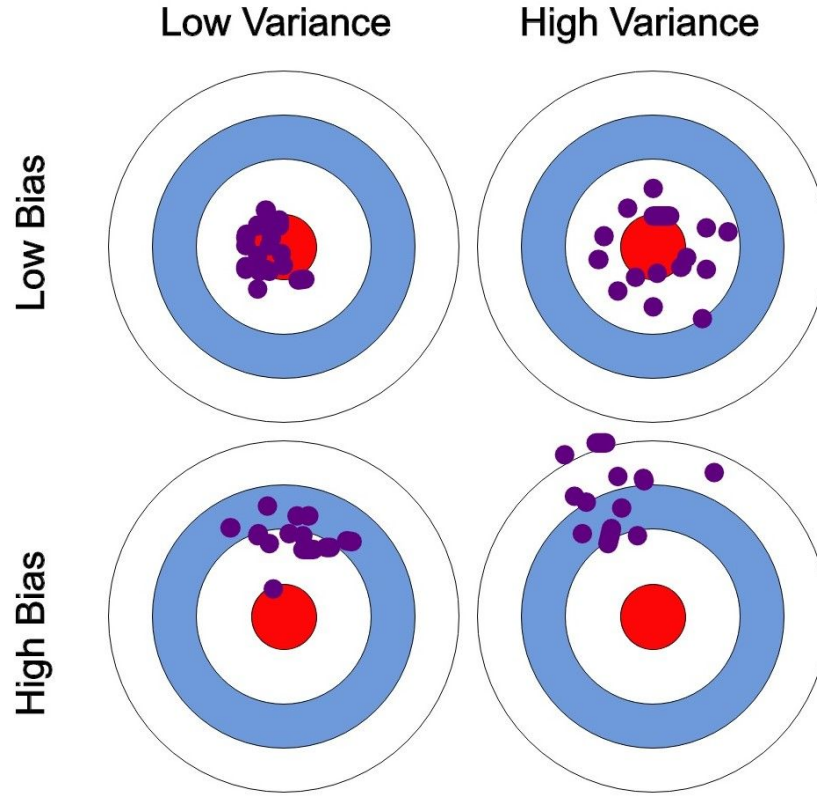
Variance control

Bagging, Boosting

Tired of having a model with a
variance too high?

Have you tried having two models?

Averaging models is called Ensembling



Where to get more models?

Bagging:

- Split data in random chunks (uniform sampling with replacement)
- Train 1 model per chunk
- Aggregate models
 - Averaging (regression)
 - Voting (classification)
 - Your call (new problem)

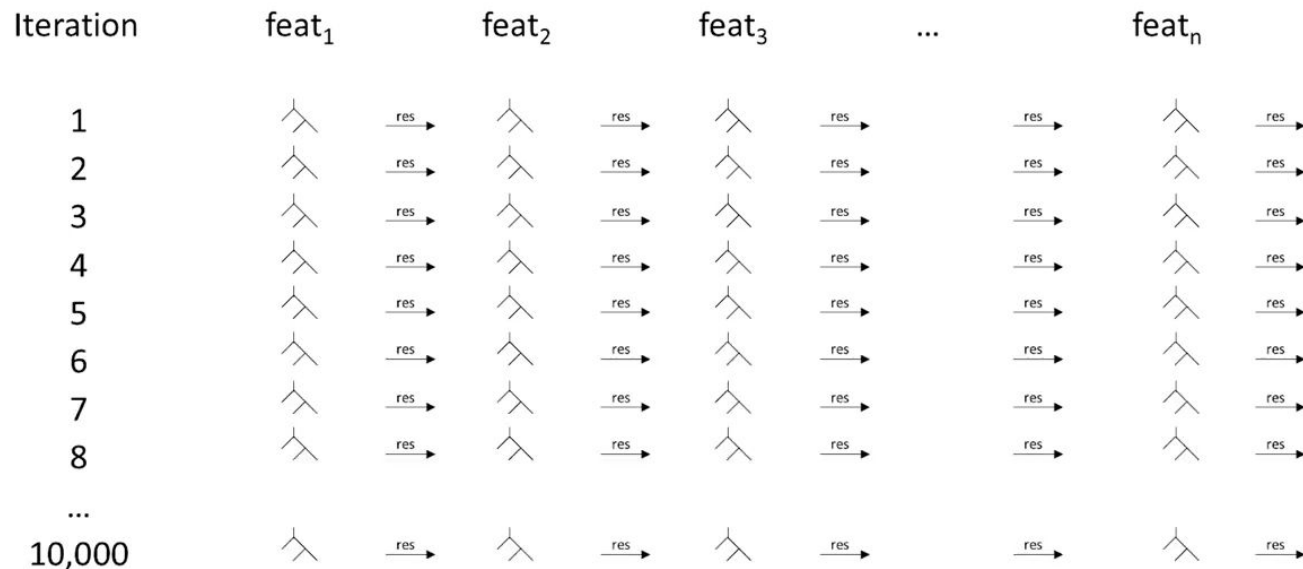
Boosting:

- Set data sampling weights equally
- Sample a chunk of data
- Train 1 model
- Reweight data sampling by mistake (failed data samples are more likely)
- Repeat until satisfied
- Aggregate models like bagging

EBMs as a practical example

[Images from Rich Caruana's original presentation](#)

EBMs are neat GAIMs

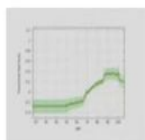
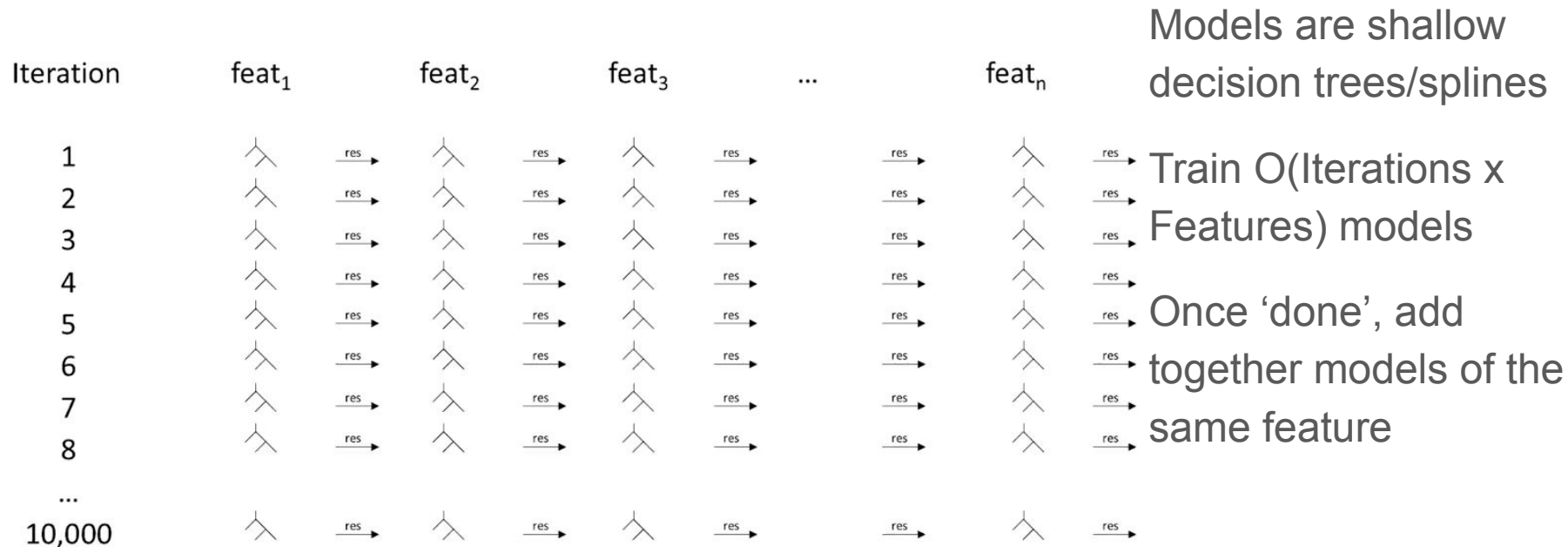


Models are shallow
decision trees

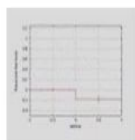
Train $O(\text{Iterations} \times \text{Features})$ models

Once 'done', add
together models of
the same feature

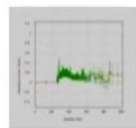
EBMs are neat GA(I)Ms



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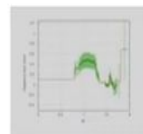
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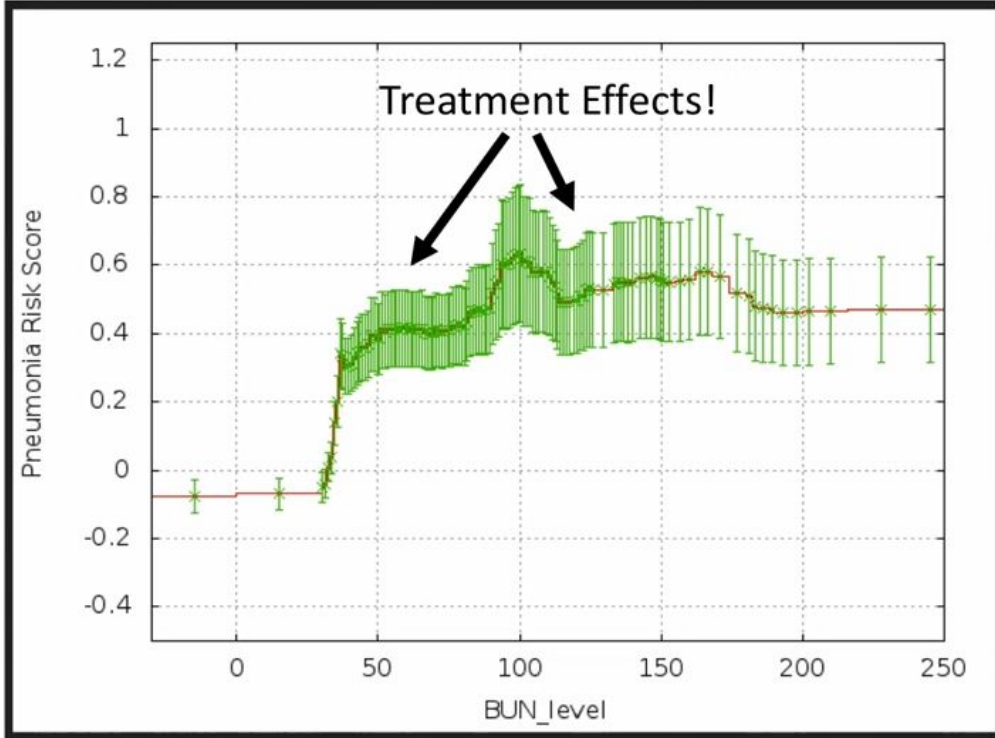
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EBM Interpretability & Usages



- Because models are trained without other variables, these are subsumed under original distribution
- That + predicting $g(y)$ means that they can be directly projected in Y space like here
- Interpretable visuals are cool (can be dangerous: pareidolia)

Joint work with Sarah Tan

Variable interaction detection: GAM-> GAIM

- Trivial to learn $f_{12}(x_1, x_2)$, but nontrivial to learn if we care about f_{12} .
 - We'd know if we trained f_{12} , but that can be expensive
 - Using *some* 'cuts' done by splines/trees in x_1, x_2 , take average of each quadrant.
 - Residual squares are used to rank strength
 - If strong enough, **then** train f_{12}

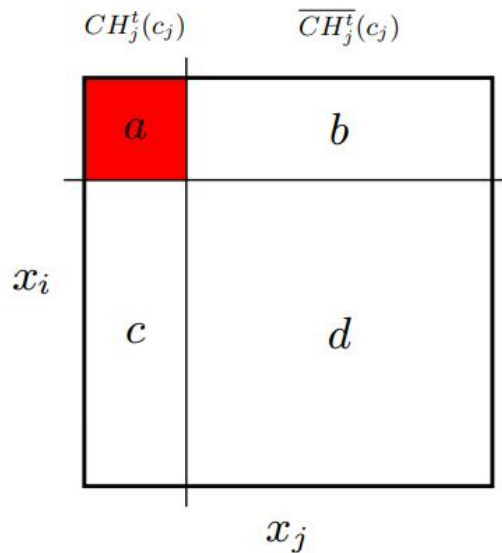


Image from Paper1

Final messages

Today we saw some models. Next week, others.

- EBMs & work shown here is cool. Remain skeptical anyway.
 - 2018 DL Earthquake predictor case: megamodel trained beaten by a traditional (basic) linear regressor.
(<https://www.nature.com/articles/s41586-019-1582-8>)
- The difference between you & a bootcamp guy is knowing the origin of stuff.
 - Nowadays it's trivial to import a library & do ML, there are millions of models.
 - Deriving which model you want to use is more important
 - Real world problems require latitude & flexibility of fundamentals, not just throwing models at the wall to see which sticks!